

1 System Model and Practical Challenges in ??

A system that takes advantage of ?? can be modeled in different ways, also depending on the use case and environment. Most layouts involve a number of ?? that transmit to a ?? or ??. The channel model is also an important aspect, as it can be affected by fading, noise, and path loss. In addition to the system model, there are also practical challenges that can affect the reliability of ??, such as synchronization errors and waveform design. To enhance the performance of ??, optimization strategies can be employed, such as power management and transceiver design.

1.a System Architecture and Channel Model

In this section the mathematical model of a network consisting of three ?? will be presented. These devices are deployed at an agricultural field and take advantage of ?? in order to transmit to the ??, by using ??. The physical environment has multipath characteristics and an impulse response with a span of three taps.

- *Modeling the Network*: Description of Edge Devices (EDs) transmitting to an Edge Server (ES)/Fusion Center (FC), possibly involving multiple antennas (??) [sahin_survey_2023](#), [yang_federated_2020](#), [hellstrom_optimal_2023](#), [cao_optimized_2022](#), [zhu_over-air_2024](#).

– **Sensor 1**: $x_1 n =$

$$2, -1, 1$$

• **Sensor 2**: $x_1 n =$

$$1, 2, -1$$

Wireless Channel Model: Consideration of channel impairments, including fading (e.g., Rayleigh fading), noise (??), and path loss [zhu_one-bit_2021](#), [evgenidis_waveform_2025](#), [cao_optimized_2022](#).

1.b Impact of Practical Impairments on OAC Reliability

1.b.i Synchronization Errors

Time, Frequency, and Phase Offsets (TO, CFO, PO): The detrimental impact of imperfect time-of-arrivals and phase alignment on signal superposition and computation accuracies [sahin_survey_2023](#), [evgenidis_waveform_2025](#), [hellstrom_optimal_2023](#).

```
1      % This is a comment
2      x = linspace(0, 2*pi, 100);
3      y = sin(x) + 0.5 * randn(size(x)); % Numbers like 0.5 will be in Math Font
4      plot(x, y);
```

1.b.ii Channel State Information (CSI) Imperfection

The need for accurate and fresh ?? at transmitters or receivers for many precoding/equalization techniques [sahin_survey_2023](#), [yang_federated_2020](#), [evgenidis_waveform_2025](#).

1.b.iii Waveform and Filter Design

Addressing time sampling error and intersymbol interference ?? inherent in digital communication transceivers when applied to OAC [evgenidis_waveform_2025](#).

1.c Optimization Strategies to Enhance OAC Performance

- *Power Management*: Strategies to mitigate aggregation errors, such as optimizing power control policies to minimize Mean Squared Error ?? or maximize convergence speed **sahin_survey_2023**, **cao_optimized_2022**, **evgenidis_waveform_2025**
- *Transceiver Design*: Joint design of pre-processing functions, precoders/beamformers, and post-processing functions (decoders) to match the channel to the desired function and minimize distortion **goldenbaum_robust_2013**, **hellstrom_optimal_2023**, **sahin_survey_2023**, **yang_federated_2020**.